

Unified Business License Data Model (DC, Chicago, NYC)

1. Purpose

This document defines a conformed (cross-city) analytics data model for business license data from Washington, DC, Chicago, and New York City. It includes an ERD description, a source-to-canonical mapping, and business rules used to normalize statuses, urgency categories, and franchise ownership proxies.

2. Source datasets

Field	Value
DC	dc_busn_lic.csv (236,746 rows; 44 columns)
Chicago	chicago_busn_lic.tsv (1,133,370 rows; 34 columns)
NYC	nyc_busn_lic.tsv (281,413 rows; 27 columns)

3. ERD description (star schema)

The model uses a star schema optimized for reporting and time-series analysis. The central fact table records one row per license record (per city/source extract). Dimensions provide conformed descriptions of businesses, locations, industries, statuses, cities, and dates.

3.1 Fact table

fact_business_license (grain: 1 row per license record)

Field	Value
Primary key	license_sk (surrogate)
Natural key	source_city + license_id
Foreign keys	city_sk, business_sk, location_sk, industry_sk, status_sk, issue_date_sk, start_date_sk, end_date_sk, status_change_date_sk
Measures	license_count (implicit), distinct_location_count (derived)

3.2 Dimensions

dim_city: city_name, state.

dim_business: legal_name, dba_name, org_id, contact details when available.

dim_location: address, street_name, zip, lat/long, and city-specific administrative geography (wards/boroughs/etc.).

dim_industry: conformed industry/license type labels; supports optional grouping.

dim_status: conformed status and status_group (Active vs Inactive).

dim_date: standard calendar attributes for time-series and seasonality analysis.

3.3 Relationships

fact_business_license joins to each dimension via surrogate keys. Each license belongs to exactly one city; typically one business and one location; one industry classification; and one current status. Date dimensions support analysis by issue/start/end/status-change dates.

Here are the exact join paths for each question,

Q1: Active vs Inactive by city & overall

FACT_BUSINESS_LICENSE.STATUS_SK → DIM_STATUS.STATUS_SK Line 2:

FACT_BUSINESS_LICENSE.CITY_SK → DIM_CITY.CITY_SK

Q2: Predominant industries

FACT_BUSINESS_LICENSE.INDUSTRY_SK → DIM_INDUSTRY.INDUSTRY_SK

Q3: Top 5 geographic areas per city

FACT_BUSINESS_LICENSE.LOCATION_SK → DIM_LOCATION.LOCATION_SK Line 2:

FACT_BUSINESS_LICENSE.CITY_SK → DIM_CITY.CITY_SK

Q4: License issuance trend over time

Line 1: FACT_BUSINESS_LICENSE.ISSUE_DATE_SK → DIM_DATE.DATE_SK

Q5: Abandonment trends

Line 1: FACT_BUSINESS_LICENSE.STATUS_SK → DIM_STATUS.STATUS_SK Line 2:

FACT_BUSINESS_LICENSE.STATUS_CHANGE_DATE_SK → DIM_DATE.DATE_SK

Q6: Seasonal variations in issuance

Line 1: FACT_BUSINESS_LICENSE.ISSUE_DATE_SK → DIM_DATE.DATE_SK

Q7: Urgency counts across cities

Line 1: FACT_BUSINESS_LICENSE.CITY_SK → DIM_CITY.CITY_SK (URGENCY is on the fact table itself — no additional join needed)

Q8: Inactive businesses with contact details

Line 1: FACT_BUSINESS_LICENSE.STATUS_SK → DIM_STATUS.STATUS_SK
Line 2: FACT_BUSINESS_LICENSE.BUSINESS_SK → DIM_BUSINESS.BUSINESS_SK
Line 3: FACT_BUSINESS_LICENSE.CITY_SK → DIM_CITY.CITY_SK

Q9: Top 10 franchise ownerships by city

Line 1: FACT_BUSINESS_LICENSE.BUSINESS_SK → DIM_BUSINESS.BUSINESS_SK
Line 2: FACT_BUSINESS_LICENSE.CITY_SK → DIM_CITY.CITY_SK (FRANCHISE_PROXY is on the fact table itself — no additional join needed)

Q10: Distribution by years since issuance

Line 1: FACT_BUSINESS_LICENSE.ISSUE_DATE_SK → DIM_DATE.DATE_SK

	From (FK)	To (PK)	Cardinality
1	FACT_BUSINESS_LICENSE.CITY_SK	DIM_CITY.CITY_SK	Many → One
2	FACT_BUSINESS_LICENSE.BUSINESS_SK	DIM_BUSINESS.BUSINESS_SK	Many → One
3	FACT_BUSINESS_LICENSE.LOCATION_SK	DIM_LOCATION.LOCATION_SK	Many → One
4	FACT_BUSINESS_LICENSE.INDUSTRY_SK	DIM_INDUSTRY.INDUSTRY_SK	Many → One
5	FACT_BUSINESS_LICENSE.STATUS_SK	DIM_STATUS.STATUS_SK	Many → One
6	FACT_BUSINESS_LICENSE.ISSUE_DATE_SK / START_DATE_SK / END_DATE_SK / STATUS_CHANGE_DATE_SK	DIM_DATE.DATE_SK	Many → One (×4 role-playing)

4. Business rules

4.1 Status mapping (to Active vs Inactive)

Statuses are normalized into status_group = Active or Inactive. The raw status is preserved.

City	Raw status examples	Normalized status_group
DC	Active, Ready to Renew	Active
DC	Abandoned, Expired, Cancelled, Revoked (and similar)	Inactive
Chicago	AAC	Active
Chicago	AAI, REV, REA, INQ (and other non-AAC codes)	Inactive
NYC	Active	Active
NYC	Inactive	Inactive

Note: Chicago publishes status as codes; if an official legend is available, update the mapping table accordingly.

4.2 Urgency mapping (renewal prioritization)

Urgency is derived for active licenses based on days until expiration (using end/expiration date):

Urgency	Rule
Immediate	days_to_expiration <= 30 (or already expired)
High	31 to 90
Medium	91 to 180
Low	> 180
Unknown	Missing expiration date

4.3 Franchise ownership proxy definition

The sources do not contain an explicit franchise indicator. For reporting, 'franchise ownership' is proxied as the number of distinct licensed locations held by the same organization within a city (organization = normalized legal_name).

Metric: franchise_proxy = COUNT(DISTINCT location_sk) per (city, business_sk).

5. Source-to-canonical mapping

A detailed column-level mapping is provided in the companion spreadsheet: Mapping_Template_Filled.xlsx.